



Engineering Project Management & Finance (UESTC 3031 & UESTCN 3012)

Design For Manufacturing (Part 2)

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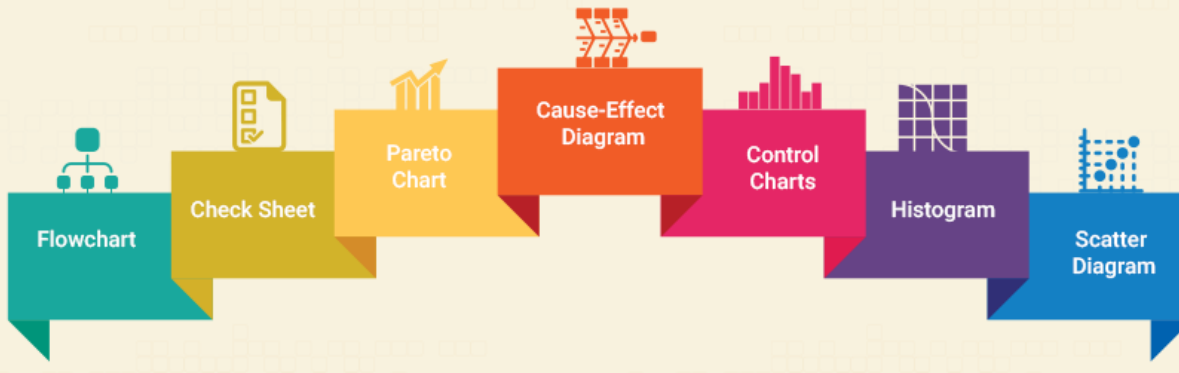
Outline

- **Part 1:** Introduction & Design for Sustainability
- **Part 2:** Quality Control & Cost of Quality
- **Part 3:** Robust Manufacturing Design
- **Part 4:** 6-Sigma & Process Capability



Quality Control (QC)

7 QUALITY CONTROL TOOLS



Definition

- Quality control (QC) is a procedure or set of procedures intended to **ensure** that a **manufactured product** or performed service adheres to a defined set of quality criteria **or** **meets the requirements** of the client or customer.
- QC is similar to, but not identical with, **quality assurance (QA)**.



The goal of QC is to **identify any defects** after a product is developed, but before it's released.

QC vs QA

- Quality assurance (QA) is a **proactive process** that focuses on **defect prevention** by progressively refining manufacturing processes, policies and procedures.
- Quality control (QC) is a **reactive process** that focuses on effectively **identifying defects** in completed products before they can be shipped to customers.

Quality Assurance Vs Quality Control

QUALITY ASSURANCE

Focus on the prevention of defects

Proactive process

Process-based approach

Manages Quality

QUALITY CONTROL

Focus on the identification of defects

Reactive process

Product-based approach

Verify the Quality



7-QC Tools

By using seven basic tools of quality, we can manage the quality of our product or process effectively. Known around the world as the **seven-quality control (7-QC) tools**, they are:

- ✓ Flowchart
- ✓ Check sheet
- ✓ Pareto chart
- ✓ Cause-and-effect diagram
- ✓ Control chart
- ✓ Histogram
- ✓ Scatter diagram



Flowchart

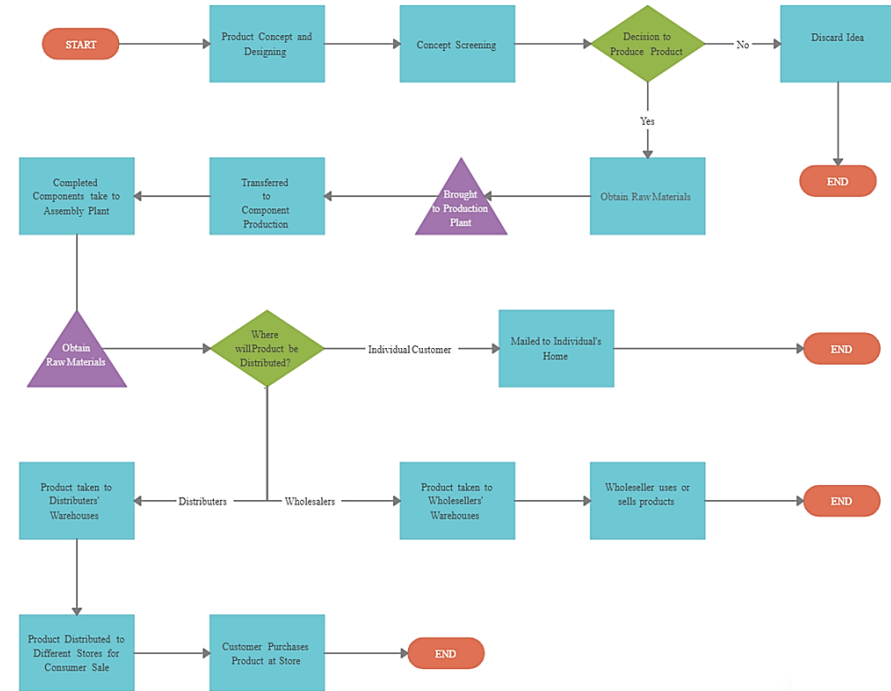
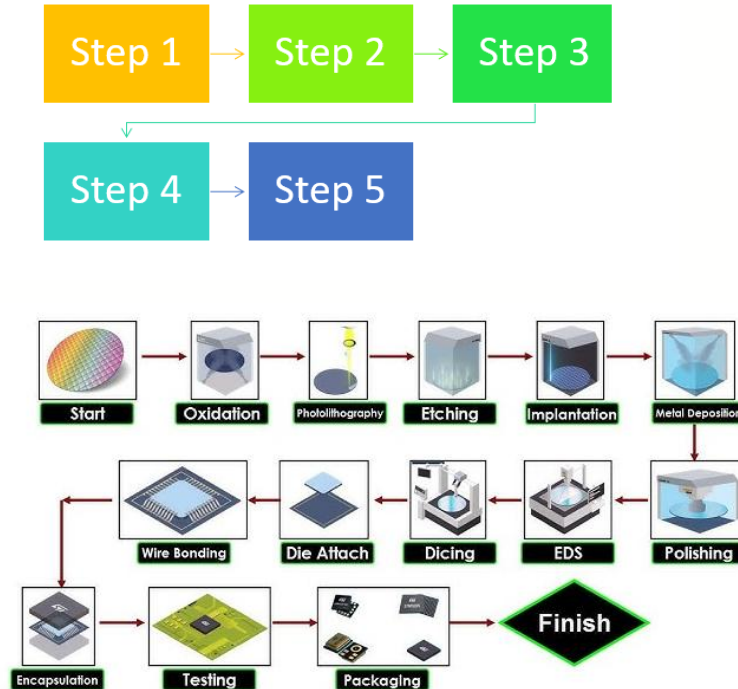
A process flowchart is a diagram that shows the **sequential steps** of a process and the decisions needed to make **the process work**. Within the chart/visual representation, every step is indicated by a shape. These shapes are connected by lines and arrows to show the movement and direction of the process.

When to use:

- ✓ To study a process for improvisation
- ✓ To document a process
- ✓ When planning a project



Flowchart (Examples)



Check Sheet

A check sheet is a structured, prepared form for **collecting and analyzing data**. This is a generic data collection and analysis tool that can be adapted for a wide variety of purposes and is considered one of the seven basic quality tools.



When to use:

- ✓ When data can be collected and observed by similar location and person
- ✓ When collecting data from a production
- ✓ When collecting data on the pattern of events, problems or defects.



Check Sheet (Examples)

Tally Sheet

Model		Part Name	Part No:
Operator		Location	Model
S.No	Defect	Frequency	Total
1	Crack		23
2	Blow hole		16
3	Wrinkle		28
4	Rough Surface		18
5	Dent		26
6	Blister		14

Project Name: _____
 Name of Data Recorder: _____
 Location: _____
 Data Collection Dates: _____

Defect Types/ Event Occurrence	Dates							TOTAL
	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	
Defect 1								
Defect 2								
Defect 3								
Defect 4								
Defect 5								
Defect 6								
Defect 7								
Defect 8								
Defect 9								
Defect 10								
TOTAL								

Check Sheet

Date:																		
Verification Stage		Section:																
Product:		Machine:																
Total inspected:		Inspector:																
Batch:		Shift:																
Specification (weight)	Variation	Verifications																Frequency
	< -0.03	X																
	-0.03	X																
	-0.02	X	X	X														
	-0.01	X	X	X	X	X	X	X	X									
5.20	0	X	X	X	X	X	X	X	X	X	X	X	X	X				
	0.01	X	X	X	X	X	X	X										
	0.02	X	X	X														
	0.03	X	X															
	> 0.03	X																
TOTAL																		

Cause-Effect Diagram

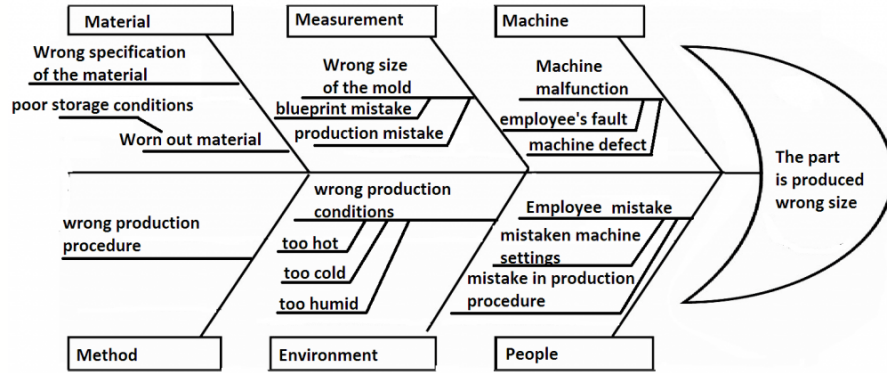
A cause-effect diagram is a visual tool used to logically organize possible causes for a specific problem or effect by graphically displaying them in increasing detail, suggesting causal relationships among theories. A popular type is also referred to as a **fishbone or Ishikawa diagram**.

When to use:

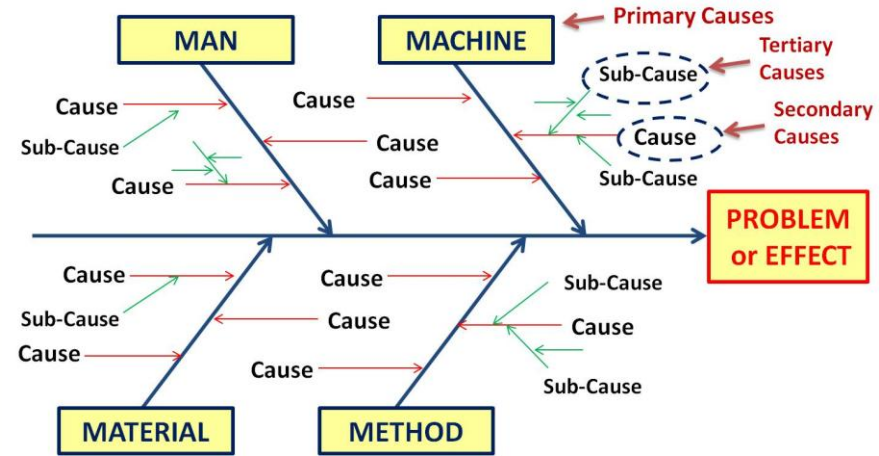
- ✓ To study a process for improvisation
- ✓ To document a process
- ✓ When planning a project



Cause-Effect Diagram (Examples)



CAUSE AND EFFECT DIAGRAM



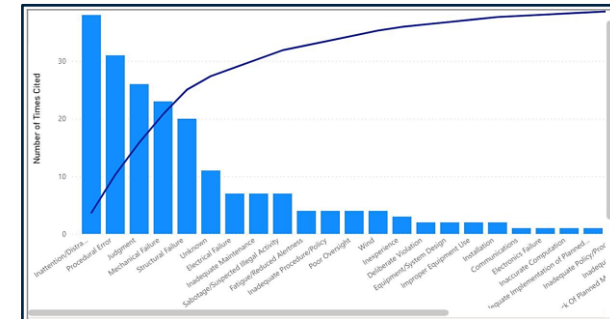
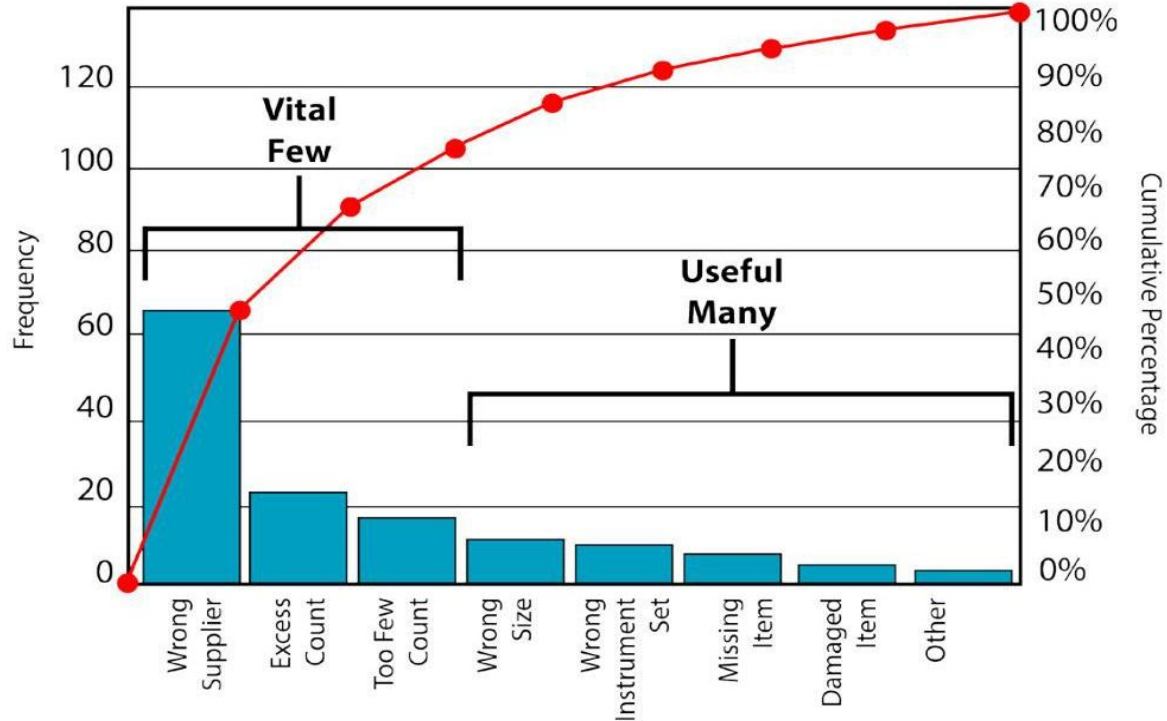
Pareto Chart

Pareto Chart is a bar chart **sorted** in descending order from the **highest frequency to the lowest frequency** from left to right. The height of the bars reflects the frequency or the impact of the problems. Based on the **Pareto principle (also known as 80/20 rule)**, which states that **80% of the effects** are caused by **20% of the causes**, the Pareto chart is used to determine the most important reasons for a certain effect.

When to use:

- ✓ When analyzing data about the frequency or cause in the process
- ✓ When there are issues or causes, and you want to focus on the important one
- ✓ When analyzing broad causes by looking at the specific parts

Pareto Chart (Example)



Control Charts

The control chart is a graph used to **study how a process changes over time**. Data are plotted in time order. It is essentially a statistical chart that helps decide if an industrial process is within control and proficient at meeting the client characterized detail limits. Control graph is also called **Shewhart Chart**.

When to use:

- ✓ To find or correct issues as they occur in an ongoing process
- ✓ When predicting the specific range of outcomes from a process
- ✓ When determining if a process is stable



Control Charts

A control chart always has a central line for the average, an upper line for the Upper Control Limit (UCL), and a lower line for the Lower Control Limit (LCL). These lines are determined from historical data. Let assume there are N historical data (sample points) from the process, then three parameters including average ($\bar{X}$), UCL and LCL are computed by:

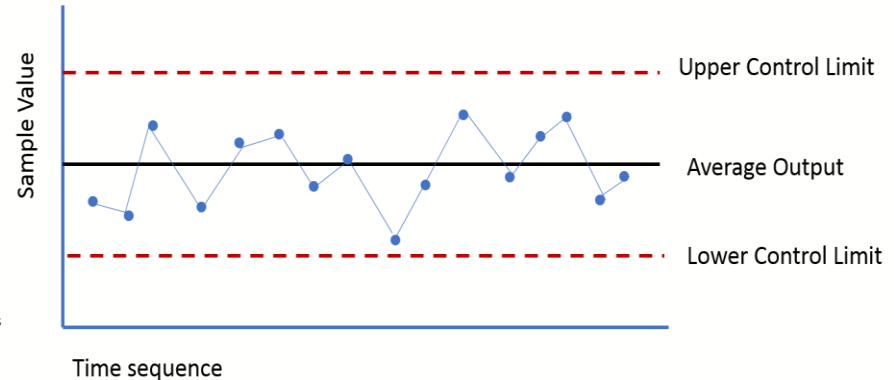
$$\bar{X} = \frac{\sum_{i=1}^N x_i}{N}$$

Where x_i shows the value of i th historical data.

$$\hat{\sigma} = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{X})^2}{N}}$$

$$UCL = \bar{X} + A\hat{\sigma}$$

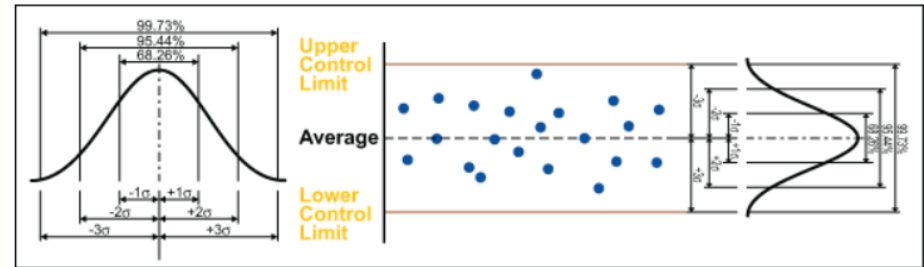
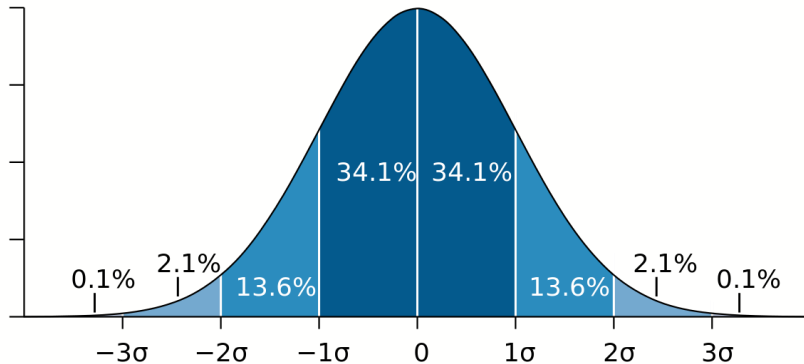
$$LCL = \bar{X} - A\hat{\sigma}$$



Computing A is discussed in next slide.

Control Charts

Control rules take advantage of the normal curve in which $A = 1$, then 68.26 percent of all data is within plus or minus one standard deviation from the average, when $A = 2$, then 95.44 percent of all data is within plus or minus two standard deviations from the average, and by $A = 3$, then 99.73 percent of data will be within plus or minus three standard deviations from the average.

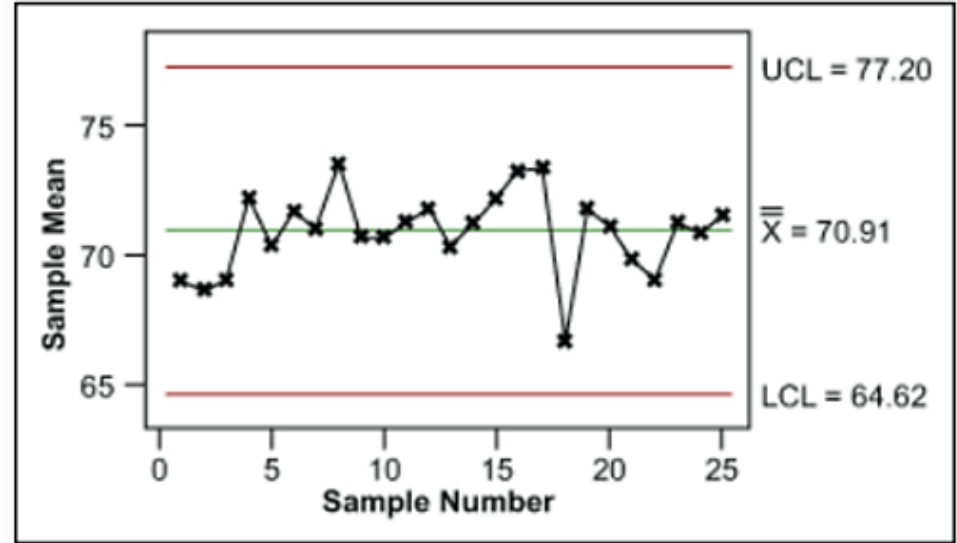


Control Charts

What do you think about this process?



Controlled Variation (Stable Process)



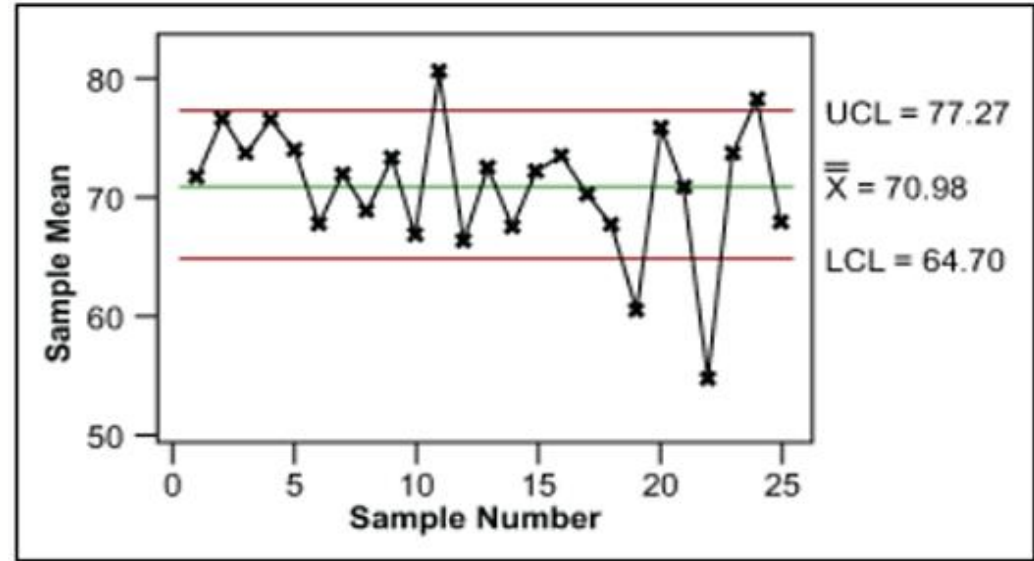
- The data points are within the control limits and there are no significant trends or patterns. This indicates that the process is stable and predictable

Control Charts

What do you think about this process?



Uncontrolled Variation (Unstable Process)



- There are several data points that are outside the control limits and there is a significant downward trend in the samples over time. This indicates that the process is not stable and there are variations in the products that are not under control.

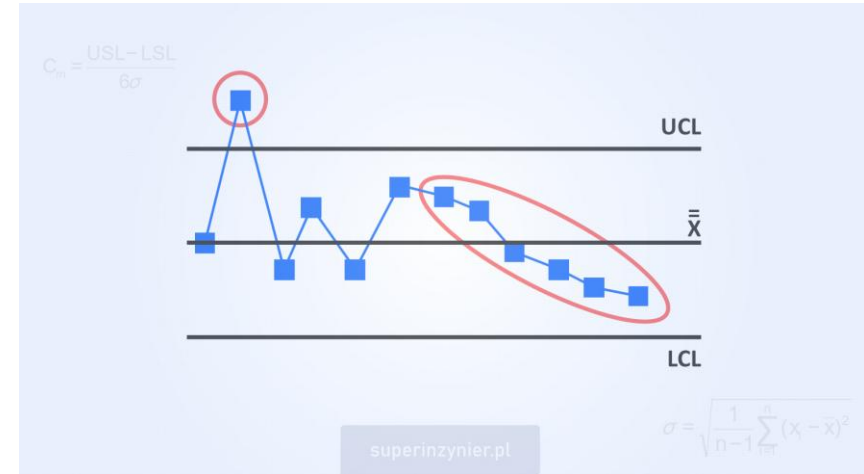
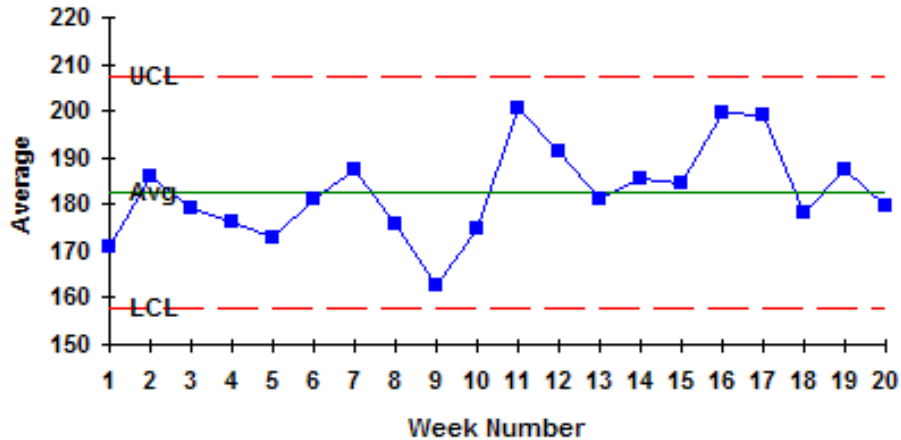
Control Charts

Non-stable control chart example (cont'd):

- **Identify the Cause:** Look for patterns or points outside control limits to understand what's causing the instability.
- **Investigate the Issue:** Check for possible factors like machine problems, material defects, or human error.
- **Implement Corrective Actions:** Fix the issue, such as repairing equipment, improving training, or changing materials.
- **Monitor the Process:** Keep using the control chart to track if the process improves and becomes stable again.
- **Verify Stability:** Ensure that the data points stay within the control limits over time, showing that the process is now stable.

Control Charts (Examples)

Xbar Chart: Bowling Scores (Avg=182.7, UCL=207.5, LCL=157.9, s=14.3, for subgroups 1-20)



Control Charts

Numerical Example

This data set represents the yield rate (%) for a particular electronic manufacturing process over 15 samples. Using this data set, a control chart can be created to monitor the process and detect any shifts or trends in the yield rate over time.

Sample Number	Yield Rate (%)
1	98.50
2	97.80
3	98.30
4	99.10
5	99.30
6	94.32
7	98.40
8	98.60
9	98.80
10	99.00
11	99.10
12	98.80
13	98.90
14	99.10
15	98.20

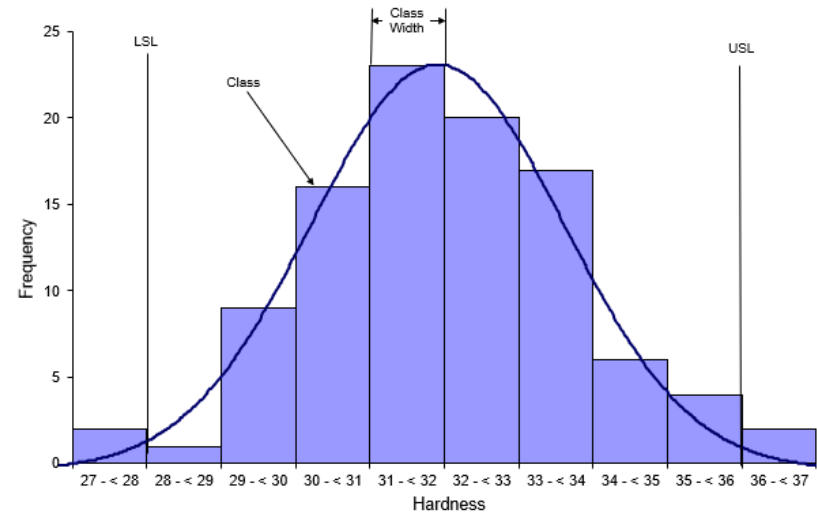
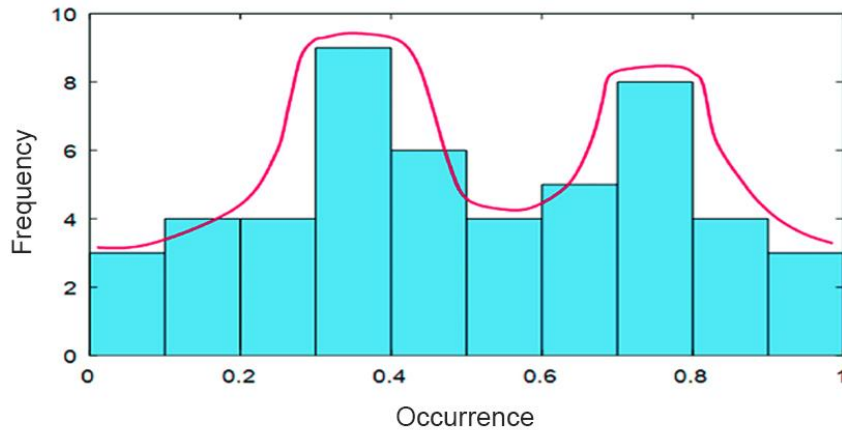
Histogram

A histogram is essentially a **bar chart** and allows you to **understand** at a glance **the variation** that exists in a process. Although, it creates a "lumpy distribution curve" that can be used to help identify and eliminate the causes of process variation.

When to use:

- ✓ When analyzing if a process can meet customer's requirements
- ✓ When analyzing the numerical data
- ✓ When determining if the outputs of two or more processes are different

Histogram (Examples)



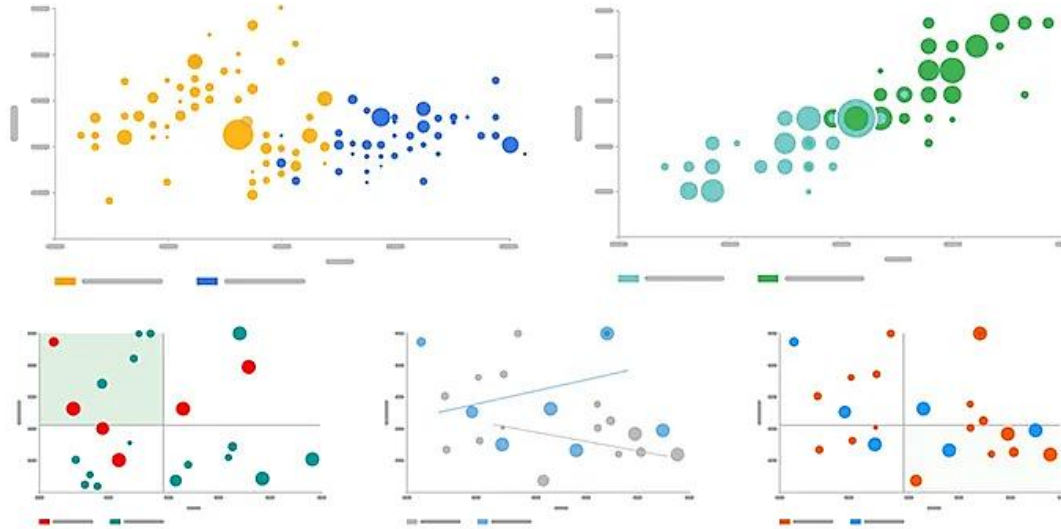
Scatter Diagram

The scatter diagram graphs pairs of numerical data, with **one variable on each axis**, to look for a **relationship** between them. If the variables are correlated, the points will fall along a line or curve. The better the correlation, the close-fitting the points will hug the line.

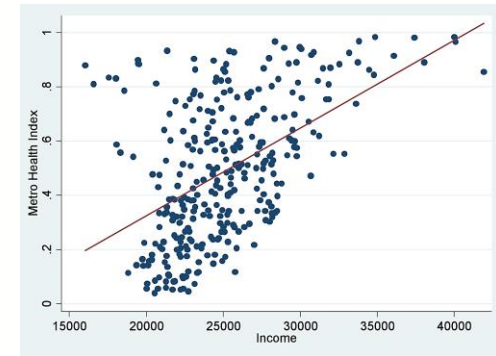
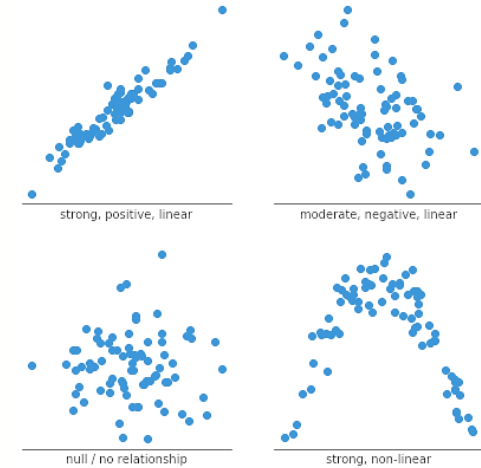
When to use:

- ✓ When you have a paired numerical data
- ✓ When testing for autocorrelation before constructing a control chart
- ✓ When trying to identify the root causes of issues

Scatter Diagram (Examples)



Top 5 Scatter Plot Examples to Get You Started with Data Visualization



Scatter Diagram

Numerical Example

This data set represents the temperature (°C) of a production line and the corresponding number of component failures during a certain period. The data can be used to create a scatter diagram to investigate the relationship between temperature and failure rate. The temperature data is plotted on the x-axis, and the number of failures on the y-axis. The resulting scatter plot can be analyzed to determine whether there is a correlation between temperature and failure rates.

Sample Number	Temperature (°C)	Failure Rate in Component A	Failure Rate in Component B
1	22	3	7
2	29	6	7.4
3	25	4	6
4	28	8	6
5	30	10	7.5
6	21	2	6.5
7	26	5	7
8	24	4	5
9	27	7	6.3
10	23	3	7

Cost of Quality (COQ)



Definition

- Armand V. Feigenbaum first described the concept of the cost of poor quality in a 1956 Harvard Business Review article.
- Cost of quality (COQ) refers to costs incurred while ensuring that you get high-quality deliverables. It also includes the cost of dealing with any defects in your work.

COSTS OF QUALITY



Quality Cost Curve

In 1999, Juran published the 5th addition of Juran's Quality Handbook where he included the following depiction of the Quality Cost Curve.



Classification

The cost of quality can be categorized into four categories:

- Prevention Cost
- Appraisal Cost
- Internal Failure Cost and
- External Failure Cost



Prevention Costs

Prevention costs associated with activities specifically designed to prevent poor quality in products. The examples of the prevention costs include:

- Quality planning
- Education and training
- Conducting design reviews
- Supplier reviews and selection
- Process planning and control
- Product modifications
- Equipment upgrades



Appraisal Costs

Appraisal costs associated with activities specifically **designed to measure, inspect, evaluate or audit** products to assure conformance to quality requirements. The examples of the appraisal costs include:

- Test and inspection (receiving, in-process and final)
- Supplier quality control
- Process Audits
- Calibration

Internal Failure Costs

Internal Failure Costs are the costs that are associated with **defects found within the organization** before the customer receives the product or service. The examples of the internal failure costs include:

- In-process scrap and rework
- Troubleshooting and repairing
- Design changes
- Inventory required to support poor process yields and rejected lots
- Re-inspection / retest of reworked items

External Failure Costs

External Failure Costs are the costs that are associated with **defects found after the customer receives the product or service**. The examples of the external failure costs include:

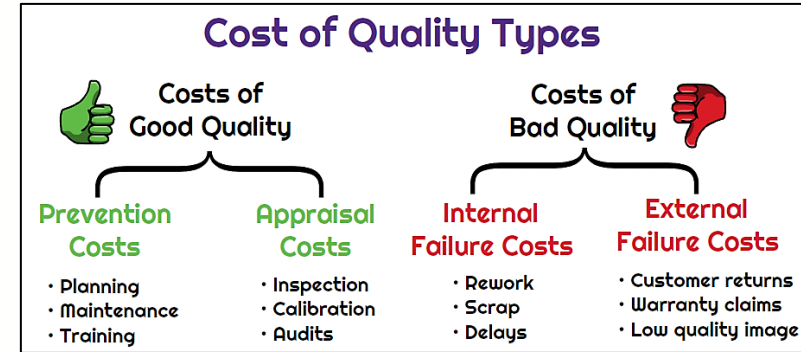
- Sales returns and allowances
- Replacing defective products
- Service level agreement penalties
- Service labor and costs incurred due to warranty obligations
- Product recalls / Legal claims
- Lost customers and opportunities
- Downgrading
- Processing of customer complaints

Examples (Classification of the Cost of Quality)

1- Cost of getting company certified to ISO 9001

2- The cost of quality-related training

3- Review of drawings



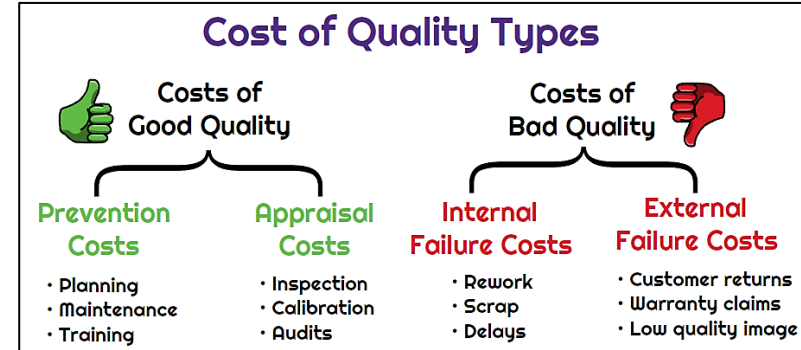
Examples (Classification of the Cost of Quality)

4- Preparation of quality management system and inspection procedures

5- Product recall and warranty repairs

6- Defects found by customers after receiving the product

7- Inspection and re-inspection of the repaired product



Visible vs Invisible Cost of Quality

COST OF POOR QUALITY

Interestingly, the hidden (invisible) costs of poor quality are often much greater than the visible costs that organizations track.

Visible

*rejection, rework, repair
cost and the cost of
inspection etc.*

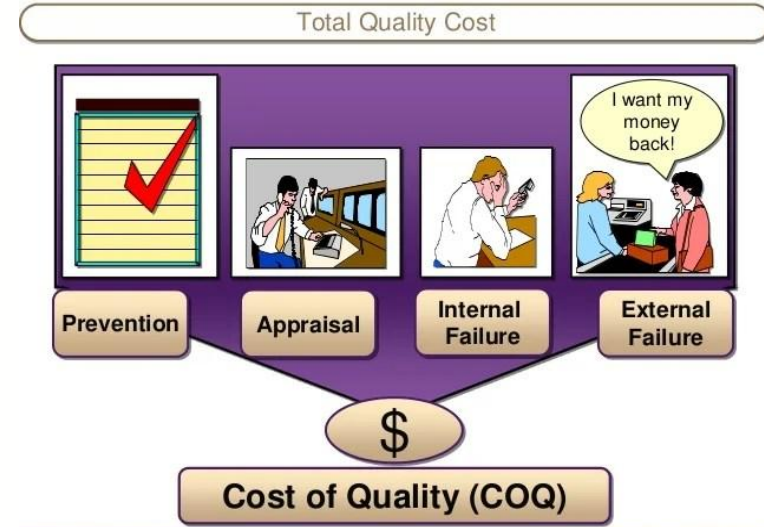
Invisible

*lost sales, excess inventory,
additional controls and
procedures, complaint
investigation, fines, legal fee etc.*



Summary

- COQ is important for companies to understand **what they need to do to reduce their overall cost of quality**. This will help them improve their business processes and operations, which ultimately lead to better profitability.
- The **worst type of cost** out of these four categories of the cost of poor quality is the **external failure costs**. Organizations should make their best effort to reduce the external failure cost.
- Spending more on prevention and appraisal costs usually leads to a reduction in internal and external failure costs.







When analyzing broad causes by looking at the specific parts, the Pareto chart is one of the applicable tools for the control of quality. Which of the following statements about a Pareto chart is true?

- A – It is a bar graph that displays the relative frequency or size of problems in descending order of importance.
- B – It is a line graph that shows the cumulative percentage of problems plotted against the total number of occurrences.
- C – It is a scatter plot that indicates the correlation between two variables.
- D – It is a pie chart that breaks down the total number of problems into categories.
- E – None of the above statements.



Which of the following statements best fits with the definition of the control charts?

- A. Control charts are used to determine whether a process is stable and predictable over time.
- B. Control charts are used to determine whether a process is profitable or not.
- C. Control charts are used to determine whether a process is meeting customer demand or not.
- D. Control charts are used to determine whether a process is efficient or not.
- E. None of the above statements is correct.



The cost of quality can be categorized into four categories: Prevention Cost, Appraisal Cost, Internal Failure Cost, and External Failure Cost. Which item below is not classified as an appraisal cost?

- A. Test and inspection (receiving, in-process, and final)
- B. Supplier acceptance sampling
- C. Product Audits
- D. Equipment upgrades
- E. Calibration



The seven basic Quality Control (7-QC) tools are widely used to manage and improve product or process quality. Which of the following is NOT one of the seven QC tools?

- A. Flowchart
- B. Pareto chart
- C. Linear programming
- D. Scatter diagram
- E. Control chart



In quality control, a cause-and-effect diagram (fishbone diagram) is mainly used to:

- A. Measure product dimensions during inspection
- B. Identify and categorize possible root causes of a problem
- C. Forecast future product demand
- D. Schedule manufacturing operations
- E. Calculate production cost



The Cost of Quality (CoQ) can be categorized into four types: Prevention Cost, Appraisal Cost, Internal Failure Cost, and External Failure Cost. Consider the following two costs in a manufacturing company: (1) cost of quality-related training for employees, and (2) product recall and warranty repairs after the product reaches customers. How should these two costs be classified?

- A. Training – Prevention Cost; Product recall – External Failure Cost
- B. Training – Appraisal Cost; Product recall – Internal Failure Cost
- C. Training – Internal Failure Cost; Product recall – Prevention Cost
- D. Training – External Failure Cost; Product recall – Appraisal Cost
- E. Training – Appraisal Cost; Product recall – Prevention Cost



A manufacturer of printed circuit boards (PCBs) wants to monitor the defect rate over time to ensure the products meet quality standards. After collecting data for a certain period, the manufacturer plots the defect rate and observes the trend over time to determine whether the process is under control or not. Which quality control tool is most appropriate for this purpose?

- A. Histogram
- B. Control Chart
- C. Check Sheet
- D. Pareto Chart
- E. Cause-and-Effect Diagram

In the Next Lesson:

Part 3: Robust Manufacturing Design

